Conjecture 00000005026: a genuine two-variable characterization

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1 The stated lower bound has an unrestricted counterexample

The source defines a "universal sentence" to mean a sentence characterizing a unique structure up to isomorphism. It specifies no target structure, signature or restricted model class, and asserts that four variables is a lower bound. In the empty first-order language with equality, consider the actual sentence

$$\varphi = \exists x \, \forall y \, (y = x).$$

It uses two bound-variable names and characterizes the one-element structure up to isomorphism. It even excludes empty structures: the existential quantifier requires an element. Thus the global four-variable lower-bound clause, under the source's provided definition, is false. We do not claim this sentence characterizes some different unspecified rich target, nor determine optimal variable requirements for a restricted encoding problem.

The word "universal" here follows the source's custom categoricity definition, not an assertion that φ is syntactically universally quantified. For that other reading, the package separately supplies $\psi = \forall x \forall y (y = x)$. Under the explicit standard *nonempty-model* convention, ψ also characterizes exactly singleton structures and uses two bound variables. Without nonemptiness, ψ also admits the empty structure; this issue is not hidden in the main witness.

2 Actual syntax, semantics and model isomorphisms

Lean uses Mathlib's actual `FirstOrder.Language.empty`, `BoundedFormula`, terms, sentences and realization semantics. The equality body is the actual AST

$$\mathrm{equal}\big(\mathrm{var}(1), \mathrm{var}(0)\big) \quad \text{in the bound-variable context } \mathrm{Fin}\, 2.$$

The sentence is formed by the actual universal and derived existential constructors. Lean proves directly from realization that a structure M satisfies φ if and only if $\exists x \in M \, \forall y \in M \, (y = x)$. The actual one-element type `Unit` is a model. Every satisfying structure has a genuine `Language.Equiv` to it, with bijection and preservation of all interpretations. The equivalence is also proved in the converse direction, and every two satisfying structures are genuinely isomorphic.

These semantic and isomorphism theorems allow arbitrary actual empty-language interpretations, not just a selected interpretation. The package also proves uniqueness of empty-language structure data on each underlying type.

3 A structural two-variable certificate

Two recursive functions inspect the actual formula AST. `binderCount` counts quantifier constructors (expanding the derived existential without adding binders), while `contextWidth` is the maximum allocated bound-variable context size among subformulas. Both equal 2 for each displayed sentence. The explicit equality-body certificate uses only the two actual indices in Fin 2. This is a sufficient two-variable certificate for these specific formulas, not a general identity between allocated contexts and minimal reusable variable width. It does not confuse a closed sentence's zero free-variable count with its bound-variable use.

The final theorem gives an actual sentence of width 2 with an actual model and universal model-isomorphism characterization, and explicitly negates the unqualified global lower bound 4. Lean 4.33.0 and pinned Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are supplied. All 13 theorems, the canonical interpretation and the proof-bearing model equivalence receive 15 transitive axiom audits, followed by full official `leanchecker --fresh Main`. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted, with separate supervising AI source/proof/ report/PDF review. This is not human certification; organizer acceptance is pending. The original bilingual source is preserved.